Open-Set Cross-Scene Hyperspectral Image Classification With Collaborative Feature and Classifier Adaptation Networks

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***Abstract*—Open-set hyperspectral image classification (HSIC) aims to accurately recognize known categories while rejecting unseen ones. However, existing methods typically assume that training and testing data follow the same distribution. Such an assumption is often violated in cross-scene applications with significant distribution shifts. In this article, we address a more practical and challenging task, termed open-set cross-scene HSIC, which requires simultaneous adaptation to scene distribution discrepancies and reliable detection of unknown categories. To this end, we propose a collaborative feature and classifier adaptation network (CF-CAN) that jointly enhances representation robustness and decision reliability under open-set cross-scene conditions. In particular, an adaptive multiscale feature fusion (AMFF) module is introduced to learn robust and transferable representations, while a class-adaptive dynamic thresholding (CADT) strategy estimates categorywise rejection boundaries for more reliable unknown detection. What's more, a self-supervised information mining (SSIM) module exploits high-confidence target-domain samples to facilitate domain alignment, improving the separability between known and unknown categories. By collaboratively optimizing feature extraction and classifier adaptation within a unified framework, CF-CAN improves both cross-scene generalization and open-set discrimination. Extensive experiments on three tasks demonstrate that CF-CAN consistently outperforms state-of-the-art HSIC methods.**

***Index Terms*—Distribution shift, hyperspectral image classification (HSIC), open-set recognition.**

I. Introduction

HYPERSPECTRAL images have garnered significant attention in remote sensing due to their rich and continuous spectral information, which enables fine-grained discrimination of ground objects [1], [2], [3]. As a fundamental task in hyperspectral image analysis, hyperspectral image classification (HSIC) has been extensively studied in recent years and successfully applied to various practical scenarios [4], [5], [6]. However, existing HSIC methods are developed under a closed-set assumption, where all test categories are presumed to belong to the label space defined during training. In practical applications, this idealized assumption is often violated due to the emergence of novel categories not seen during training. This gap has motivated increasing attention on open-set recognition in HSIC.

To address the interference caused by unseen categories, open-set HSIC has emerged as a significant research direction. The goal is to accurately recognize known categories while effectively rejecting unknown ones. Existing open-set HSIC methods primarily focus on enhancing feature discriminability or refining rejection boundaries to increase the separation between known and unknown categories in the feature space. For instance, Sun et al. [7] proposed a spectral–spatial MLP-like network with reciprocal points learning for open-set HSIC, achieving improved open-set classification through more refined decision boundaries. Qin et al. [8] introduced a language-enhanced dual-level contrastive learning network for open-set HSIC, which improves feature discriminability by incorporating textual supervision and dual-level visual-language contrastive learning. Although these proposed methods have achieved promising results, their strong performance typically relies on an idealized independent and identically distributed assumption. Under this assumption, both training and testing data are presumed to be drawn from the same distribution. However, hyperspectral images collected in real-world scenarios often exhibit substantial distribution shifts across different scenes. Such shifts mainly arise from differences in sensor characteristics, illumination conditions, and geographic environments, making it difficult for a model trained on one scene to generalize directly to another.

In recent years, extensive research has been conducted on cross-scene HSIC, primarily from the perspectives of domain generalization, cross-domain few-shot learning, and unsupervised domain adaptation. For example, Gao et al. [9] proposed a distribution-independent domain generalization method for multisource remote sensing classification involving hyperspectral imagery, improving model generalization across different data distributions. Cai et al. [10] developed a multilevel unsupervised domain adaptation framework that

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mitigates cross-scene domain discrepancies through adaptation at the image, feature, and logic levels. Li et al. [11] introduced a semantic-guided prototype learning method for cross-domain few-shot HSIC, which integrates semantic information into prototype learning to improve prototype discrimination under limited supervision in the target domain. Although existing cross-scene HSIC methods are effective in reducing domain shifts across different scenes, these methods still operate under a closed-set assumption, where the source and target domains share the same label space. Consequently, in open-set scenarios, forcing unknown target samples to align with known source categories leads to severe negative transfer. Such misalignment increases the likelihood that unknown samples are misclassified as belonging to known categories.

These limitations suggest that neither conventional open-set HSIC methods nor existing cross-scene HSIC methods are adequate for real-world scenarios involving the coexistence of unknown categories and domain shifts. On the one hand, conventional open-set HSIC methods misclassify shifted known samples as unknown or fail to detect true unknown samples. On the other hand, existing cross-scene HSIC methods incorrectly align unknown target samples with known source categories, leading to degraded classification performance. Although open-set domain adaptation (OSDA) has been extensively studied for conventional RGB image classification, directly applying existing OSDA methods to open-set cross-scene HSIC remains challenging. The difficulty stems from the intrinsic properties of hyperspectral imagery: high-dimensional spectral bands, strong spectral–spatial correlations, significant scene-dependent spectral shifts within the same category, and highly similar responses across different categories. These factors collectively increase the risk of misclassifying unknown categories as known ones, rendering existing OSDA methods ill-suited for the HSIC task. Therefore, a unified framework for open-set cross-scene HSIC is needed to simultaneously address cross-scene domain shifts and enable reliable discrimination between known and unknown categories.

To address the above challenge, several OSDA methods have been developed for open-set cross-scene HSIC. Representative approaches combine reliable target sample selection, cross-domain alignment, and discrimination between known and unknown categories through joint or decoupled strategies. Despite these advances, an important question is how to establish explicit feedback between feature representation learning and classifier adaptation. These two processes are inherently interdependent. Discriminative and transferable feature representations yield more reliable classifier predictions, whereas reliable classifier decisions facilitate the selection of informative target samples. These selected samples, in turn, provide target-domain supervision for refining both the feature extractor and classifier. Establishing such a mutually reinforcing loop can improve both cross-scene generalization and discrimination between known and unknown categories, motivating the collaborative adaptation design of a collaborative feature and classifier adaptation network (CF-CAN).

To address the aforementioned issues, we propose a CF-CAN for open-set cross-scene HSIC. The CF-CAN jointly addresses cross-scene domain shifts and open-set recognition through robust feature-level representation learning and classifier-level dynamic threshold adjustment. In particular, we introduce an adaptive multiscale feature fusion (AMFF) module to capture deep interactions among multiscale spectral–spatial features. The AMFF module adaptively integrates the complementary information of these features, thereby enhancing the robustness of the learned representations. We further propose a class-adaptive dynamic thresholding (CADT) strategy, which independently estimates rejection thresholds for each category to mitigate misclassification and stabilize the decision boundaries between known and unknown categories. To strengthen domain alignment and improve the separability between known and unknown categories, we design a self-supervised information mining (SSIM) module that generates high-confidence pseudolabels from target-domain data, providing informative supervision. By collaboratively optimizing these components within a unified end-to-end framework, CF-CAN enables joint adaptation of the feature extractor and classifier, thus enhancing cross-scene generalization and enabling reliable rejection of unknown categories in open-set scenarios.

The main contributions are outlined as follows.

1) We propose a novel CF-CAN framework for open-set cross-scene HSIC from the perspective of collaborative feature and classifier adaptation. This work reveals that the lack of effective coordination among feature adaptation, unreliable rejection boundaries, and insufficient target data utilization severely compromises the ability to reliably distinguish known from unknown categories.
2) We design an AMFF module to learn robust and transferable representations, a CADT strategy to stabilize classwise rejection boundaries, and an SSIM module to exploit informative supervision from target-domain data. Reliable target-domain supervision bridges feature representation learning and classifier adaptation, enabling the two processes to reinforce one another, improving cross-scene generalization and discrimination between known and unknown categories.
3) Extensive experiments on three benchmarks demonstrate the superiority of the proposed CF-CAN over existing open-set HSIC and cross-scene HSIC methods, with stronger generalization capability on the open-set cross-scene HSIC task.

II. Related Works

A. Open-Set HSIC

Open-set HSIC aims to accurately classify known land-cover categories while rejecting unknown categories that are not present during training but appear in complex remote sensing environments. Existing open-set HSIC methods can be broadly grouped into two categories: representation learning-based open-set HSIC methods and rejection calibration-based open-set HSIC methods.

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For representation learning-based open-set HSIC methods, the key objective is to improve the separability between known and unknown categories by learning more compact and discriminative feature structures. For example, Xie et al. [12] proposed a feature consistency-based prototype network to refine feature representations with prototype-guided learning. Ji et al. [13] developed a spectral–spatial evidential learning network to model evidential uncertainty and improve open-set discrimination. Although these methods have achieved promising performance, they mainly focus on improving feature discriminability under the assumption that the training and testing data follow the same distribution. They do not explicitly consider severe distribution shifts across different scenes.

Rejection calibration-based methods, by contrast, focus on refining decision criteria to ensure the reliable rejection of unknown samples. For instance, Duan et al. [14] developed a semisupervised open-set HSIC method, which incorporates threshold selection and statistical modeling to improve unknown-category rejection. Ji et al. [15] proposed a global distribution-aware network to capture global data distribution and intraclass multimodal structure, improving the performance of open-set HSIC. However, these methods still produce unreliable confidence scores and rejection boundaries under severe distribution shifts across different scenes.

Despite significant progress in open-set HSIC, the effectiveness of current methods still heavily relies on the idealized assumption that the training and test data follow the same distribution. In practice, hyperspectral images collected in real-world scenarios often exhibit substantial distribution shifts. Under such circumstances, existing methods often struggle to simultaneously maintain robust recognition of known categories and reliable rejection of unknown categories. Consequently, there remains considerable potential for further exploring open-set cross-scene HSIC to address this challenge.

B. Cross-Scene HSIC

In recent years, cross-scene HSIC has emerged as an important task to improve model generalization across different scenes. Existing cross-scene HSIC methods can be broadly classified into three paradigms: domain generalization, cross-domain few-shot learning, and unsupervised domain adaptation.

Domain generalization-based methods improve robustness to unseen target scenes without relying on any target-domain data during training. For instance, Zhang et al. [16] developed a single-source domain expansion network to enrich source-domain diversity, improving generalization to unseen scenes. Li et al. [17] introduced a counterfactual-augmented domain generalization framework to enhance model robustness via causal disentanglement and counterfactual generation. These methods are particularly practical when target-domain data are limited or unavailable. However, they mainly focus on learning domain-invariant representations and do not explicitly address reliable discrimination between known and unknown categories under open-set conditions.

Cross-domain few-shot learning methods facilitate rapid adaptation to new scenes with only a few labeled target samples. For example, Gui et al. [18] proposed a cross-domain few-shot HSIC method based on circle loss and dense graph convolution, which improves discriminative prototype learning under severe sample scarcity. Paeedeh et al. [19] introduced a mixup foundation model that leverages a pretrained remote sensing foundation model, coalescent projection, and mixup domain adaptation to improve adaptation under domain discrepancy. However, these methods rely on a few labeled target samples, which restricts their applicability in scenarios where target-domain labels are scarce or unavailable.

Unsupervised domain adaptation-based methods mitigate the discrepancy between labeled source domain data and unlabeled target-domain data through feature alignment, self-training, and classifier adaptation. For instance, Ding et al. [20] proposed a consistency-aware customized learning framework for cross-scene HSIC by integrating domain-level alignment with class-level contrastive learning guided by prototype consistency. Qiu et al. [21] introduced a domain fusion contrastive learning method to fuse interdomain and intradomain features, thereby improving feature discrimination. Although these methods are effective for cross-scene HSIC, they generally assume that the target domain contains only known categories, a premise that limits their use in more complex scenarios.

Despite significant progress, existing cross-scene HSIC methods largely rely on the assumption that training and test data follow the same category distribution. Nevertheless, in real HSIC scenarios, unknown categories that are absent during training often appear in the target scene. This mismatch makes conventional cross-scene HSIC methods inadequate for open-set recognition, thereby motivating the study of open-set cross-scene HSIC.

C. OSDA HSIC

To jointly address cross-scene distribution shifts and unknown categories in the target domain, recent studies have proposed various OSDA methods for open-set cross-scene HSIC. Based on their principal adaptation mechanisms, representative approaches can be analyzed along three complementary dimensions: target sample reliability modeling, cross-domain feature alignment, and discrimination between known and unknown categories. For example, Bi et al. [22] proposed a weighted generative adversarial network with dynamic thresholding, which estimates target sample reliability and adaptively weights samples during domain alignment to mitigate negative transfer. Li et al. [23] developed a pixel-aware OSDA framework, which alleviates patch-level feature bias by modeling neighborhood-level pixel uncertainty and decoupling the alignment of known and unknown features. Deng et al. [24] introduced a joint distribution matching framework with unknown-class uncertainty suppression, which reduces the distribution discrepancy between source and target known category features while suppressing the uncertainty associated with unknown categories. More recently, Liu et al. [25] proposed OSUSE, which explores unknown samples and refines target pseudolabels to improve OSDA. Liu et al. [26] introduced SoDa^2, a single-stage OSDA method that decouples the alignment of spectral and spatial representations and employs a dual-branch architecture to discriminate between known and unknown categories.

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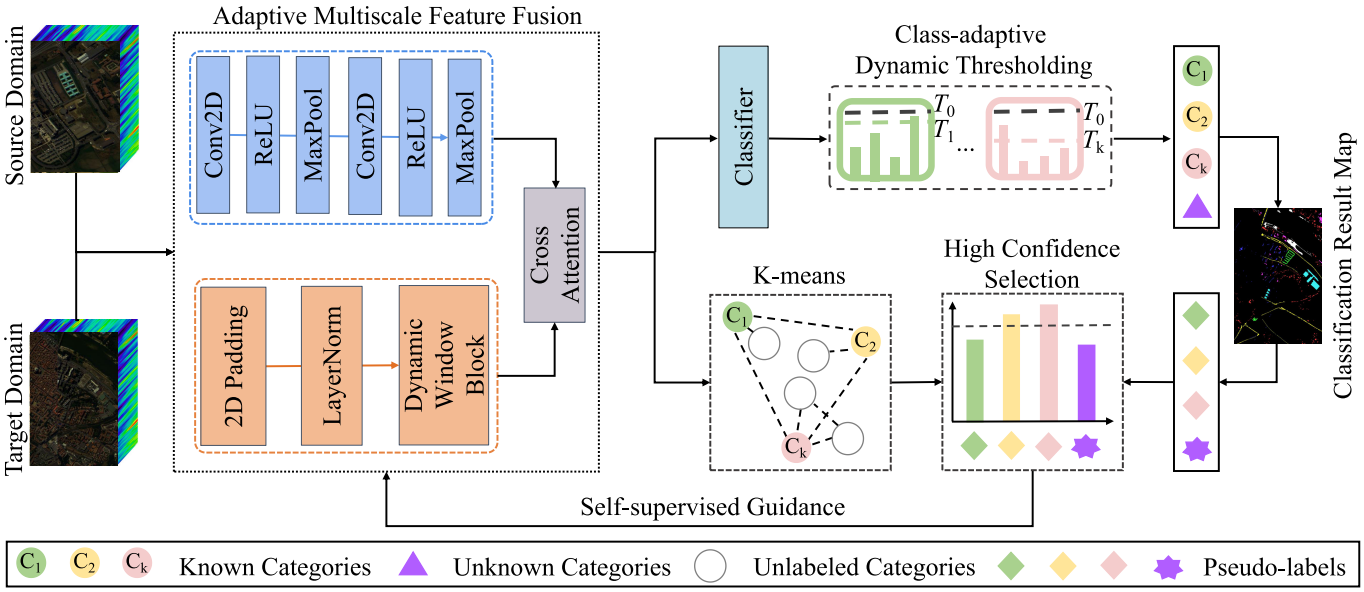


Fig. 1. Diagram illustrating the proposed CF-CAN framework.

In summary, these methods have advanced open-set cross-scene HSIC by integrating reliable target sample selection, cross-domain feature alignment, and known–unknown discrimination in different ways. Despite this progress, an important question is how to establish explicit feedback between feature representation learning and classifier adaptation. Discriminative and transferable feature representations can yield more reliable classifier predictions, whereas reliable classifier decisions facilitate the selection of informative target samples. These selected samples, in turn, provide pseudolabel-based target-domain supervision for refining both the feature extractor and classifier. Accordingly, the motivation of CF-CAN lies not in merely combining domain adaptation with open-set recognition, but in explicitly organizing feature learning, classifier adaptation, and reliable target sample selection into a mutually reinforcing loop.

III. Collaborative Feature and Classifier Adaptation Networks

The proposed CF-CAN framework comprises three core components integrated into a unified end-to-end architecture. These components work synergistically to enhance representation robustness and classifier reliability for open-set cross-scene HSIC. In particular, CF-CAN first employs the AMFF module to learn robust and transferable spectral–spatial representations from hyperspectral data, thereby improving the adaptability of the feature backbone. The AMFF module adopts a dual-branch architecture consisting of a convolutional neural network (CNN) branch and a dynamic Window-Transformer branch, which work together to capture complementary local and global features. A bidirectional cross-attention fusion mechanism is further introduced to enable deep cross-branch interaction and adaptive feature aggregation. Next, the CADT strategy serves as the core component for classifier adaptation. This strategy adaptively estimates class-specific rejection thresholds to calibrate decision boundaries, thereby reducing misclassification errors and improving the discrimination between known and unknown categories. Finally, the SSIM module is designed to exploit informative supervision from target-domain data. By generating high-confidence pseudolabels, it refines domain alignment and further strengthens the separation between known and unknown categories. In this process, the learned feature representations support more reliable classifier decisions and target sample selection, while the resulting target-domain supervision feeds back to refine both the feature extractor and classifier. Consequently, the three components are coupled within a mutually reinforcing adaptation process rather than functioning as independent modules. The overall architecture of the CF-CAN framework is illustrated in Fig. 1. Detailed descriptions of each component are provided in Sections III-A–III-C.

A. AMFF Module

In open-set cross-scene HSIC, feature representation learning is challenged by distribution shifts across scenes and interference from unknown categories. Existing methods struggle to jointly capture local spectral–spatial details and global contextual semantics, while naive fusion strategies fail to effectively integrate such complementary information. These limitations lead to insufficient representation robustness, reduced intraclass compactness, and poor separability between known and unknown categories.

To address these limitations, we propose an AMFF module for learning robust and discriminative spectral–spatial representations in open-set cross-scene HSIC. The AMFF module adopts a dual-branch architecture, where a CNN branch captures fine-grained local details and a dynamic Window-Transformer branch models long-range contextual dependencies. To effectively integrate their complementary strengths, we design a bidirectional cross-attention fusion

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mechanism that enables deep cross-branch interaction and adaptive feature aggregation. This design facilitates more discriminative representations with improved intraclass compactness and enhanced separability between known and unknown categories.

Let $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$ denote an input hyperspectral image patch, where B, C, H, and W represent the batch size, number of spectral channels, height, and width, respectively. The AMFF module extracts features through the CNN branch and the dynamic Window-Transformer branch in parallel. In the CNN branch, two convolutional layers are applied, each followed by ReLU activation and max-pooling. These layers are employed to learn local spectral–spatial representations. The resulting features are flattened and projected into a compact embedding space, yielding the local representation $\mathbf{F}_{\text{cnn}}$. This branch focuses on preserving local textures, structural boundaries, and fine-grained discriminative information.

Meanwhile, the dynamic Window-Transformer branch is used to capture multiscale contextual dependencies. First, the input patch $\mathbf{X}$ is converted into a sequence of patch tokens through a convolutional patch embedding layer with learnable positional encodings. The resulting token sequence is then processed by four dynamic window blocks. Each block integrates dynamic window attention with a convolution-enhanced spatial refinement unit. The first and third blocks perform attention within nonshifted windows, whereas the second and fourth blocks adopt shifted windows to enhance information interaction across adjacent windows. This alternating strategy is particularly well suited to hyperspectral images, as land-cover regions typically exhibit strong local spatial continuity and often extend across predefined window boundaries. Nonshifted blocks efficiently capture local spectral–spatial dependencies, whereas shifted blocks promote information exchange between adjacent windows. Consequently, alternating these two types of blocks allows information to propagate progressively across windows while preserving the computational efficiency of local attention.

Unlike conventional fixed-window attention, the proposed dynamic window attention adaptively determines the effective window size based on multiple candidate scales. Let $\{r_m\}_{m=1}^{M}$ denote the candidate scale factors. In our implementation, three candidate scale factors are considered, i.e., $\{1/2, 1, 2\}$. With respect to the base window size, the factors $1/2$, 1, and 2 define smaller, reference, and larger windows, which, respectively, capture local fine-grained patterns, base scale information, and broader spatial context. By integrating these candidate scales, the model can adapt its receptive field to diverse spatial structures without being restricted to a fixed window size. Moreover, this compact scale set strikes a good balance between flexibility and efficiency, avoiding the redundancy and extra computation incurred by using many more scales. To adaptively combine different scale factors, a lightweight predictor estimates their relative importance and generates the corresponding normalized weights $\{\alpha_m\}_{m=1}^{M}$. The dynamic window size is then determined as

$$w_d = \sum_{m=1}^{M} \alpha_m r_m w \tag{1}$$

where w denotes the base window size. In practice, w_d is further adjusted according to the current feature resolution. This adjustment ensures that the resulting window partition remains valid and compatible with the spatial dimensions of the feature map. Once the dynamic window size w_d is determined, the shift size is set to $s_d = \lfloor w_d/2 \rfloor$ for shifted blocks and $s_d = 0$ for nonshifted blocks. The floor operation ensures an integer-valued shift when w_d is odd. By maintaining an approximately half-window displacement as w_d varies, this scale-aware formulation enables consistent cross-window interaction across different receptive-field configurations. After the window partition, multihead self-attention is performed within each local window

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}}\right)\mathbf{V} \tag{2}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ denote the query, key, and value matrices, respectively, and d is the feature dimension of each attention head. For the shifted-window blocks, the feature map is cyclically shifted before window partitioning and restored after attention computation. This approach extends the effective receptive field across adjacent windows while preserving the efficiency of local attention.

To further strengthen local spatial modeling, a convolution-based spatial refinement unit is introduced after each attention layer. Let $\mathbf{Z}$ denote the token features obtained after dynamic window attention. The refinement unit first reshapes $\mathbf{Z}$ into a 2-D feature map and then injects spatial inductive bias through depthwise and pointwise convolutions. Subsequently, an MLP with a residual connection is applied to refine the features

$$\widetilde{\mathbf{Z}} = \mathbf{Z} + \mathcal{C}(\mathbf{Z}) \tag{3}$$

where $\mathcal{C}(\cdot)$ represents the convolution-enhanced self-calibration operation. By combining dynamic window attention with convolution-enhanced spatial refinement, the dynamic Window-Transformer branch can simultaneously capture long-range contextual dependencies and local structural details. After the stacked dynamic window blocks, the output features are normalized and globally pooled, resulting in the contextual representation $\mathbf{F}_{\text{wt}}$.

Although the two branches capture complementary information, a direct combination fails to fully exploit their interaction. Therefore, we introduce a bidirectional cross-attention fusion mechanism. In particular, the features from both branches are first projected into a common embedding space. Bidirectional cross-attention is then performed to enable mutual guidance between the two branches. In this way, local structural cues and global contextual semantics can be more effectively integrated. The attention-enhanced features from the CNN branch and the dynamic Window-Transformer branch are denoted as $\mathbf{F}_{\text{cnn}}^{\text{att}}$ and $\mathbf{F}_{\text{wt}}^{\text{att}}$, respectively. The final fused representation is obtained as

$$\mathbf{F}_{\text{fuse}} = \beta_{\text{cnn}}\mathbf{F}_{\text{cnn}}^{\text{att}} + \beta_{\text{wt}}\mathbf{F}_{\text{wt}}^{\text{att}} \tag{4}$$

where β_{cnn} and β_{wt} are learnable fusion coefficients. Through this design, the proposed AMFF module yields more robust and discriminative representations for open-set cross-scene HSIC. It provides a more reliable and transferable representation space for subsequent classifier adaptation.

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B. CADT Strategy

Although the AMFF module improves the robustness of feature representations, relying solely on feature learning is insufficient to achieve reliable open-set cross-scene HSIC. Beyond robust representations, dependable discrimination between known and unknown categories also requires well-calibrated decision boundaries at the classifier level. Due to cross-scene distribution shifts, different known categories often exhibit distinct confidence distributions. As a result, a single global threshold cannot provide stable separation between known and unknown samples. In addition, a fixed threshold may either reject challenging known samples or accept confusing unknown ones. To address this issue, we introduce a CADT strategy. CADT assigns an independent rejection threshold to each category and performs categorywise boundary calibration directly at the classifier level.

Given the learned feature representation $\mathbf{f}_i$ of sample $\mathbf{x}_i$, the classifier produces a category probability vector

$$\mathbf{p}_i = g(\mathbf{f}_i) = \left[p_i^{(1)}, p_i^{(2)}, \ldots, p_i^{(C)}\right] \tag{5}$$

where C denotes the number of known categories and $p_i^{(c)}$ is the predicted probability that sample $\mathbf{x}_i$ belongs to category c. Instead of employing a unified threshold for all categories, CADT maintains a categorywise threshold τ_c for each known category c. Let $\hat{y}_i$ be the predicted category corresponding to the maximum confidence. A target-domain sample is regarded as a classifier-level known candidate only if its maximum confidence exceeds the threshold associated with the predicted category, i.e.,

$$p_i^{(\hat{y}_i)} > \tau_{\hat{y}_i}. \tag{6}$$

Otherwise, the sample is treated as a candidate unknown.

During training, the categorywise decision boundaries are calibrated adaptively using confidence statistics. In particular, the threshold for class c is estimated as

$$\tau_c = \bar{p}_c - \gamma\sigma_c \tag{7}$$

where $\bar{p}_c$ and σ_c denote the mean and standard deviation of the confidence scores of target-domain samples currently predicted as category c and γ is a scaling factor. In this manner, CADT performs category-specific boundary calibration on the classifier outputs. It alleviates the limitations of a shared global threshold and improves the reliability of discrimination between known and unknown categories.

C. SSIM Module

While CADT improves categorywise boundary calibration at the classifier level, effective target-domain adaptation still requires richer supervision from unlabeled target samples. To address this, we introduce the SSIM module. Building upon CADT's classifier-level calibration, SSIM refines pseudolabel assignment by jointly leveraging classifier confidence, category-specific thresholds, and feature-space clustering. This joint strategy provides more reliable supervision from the target domain, thereby facilitating domain alignment and strengthening the separation between known and unknown categories.

To jointly exploit classifier calibration and feature-space structure, we first define an unknownness score for each target-domain sample. Given the target-domain sample set $\mathcal{T}$, the unknownness score of sample $\mathbf{x}_i \in \mathcal{T}$ is computed as

$$u_i = \tau_{\hat{y}_i} - p_i^{(\hat{y}_i)} \tag{8}$$

where $\hat{y}_i$ is the predicted known-category label. A larger u_i indicates that $\mathbf{x}_i$ is more likely to belong to the unknown category. Based on these unknownness scores, the top-R target-domain samples are selected as unknown candidates

$$\mathcal{U} = \left\{\mathbf{x}_i \in \mathcal{T} \,\middle|\, u_i \text{ is among the top-}R \text{ largest}\right\}. \tag{9}$$

Here, R is a fixed hyperparameter that determines the size of the initial unknown candidate pool rather than estimating the actual number of unknown target samples.

To further refine pseudolabel assignment, we incorporate both the known category structure and the identified unknown candidates. In particular, we perform $(C+1)$-way clustering on the target-domain features. The first C cluster centroids are initialized by the source prototypes of known categories, while the additional centroid for the unknown candidate cluster is initialized by the average feature of $\mathcal{U}$. All target-domain features are then partitioned into $(C+1)$ clusters, and each target-domain sample receives a clustering label $\hat{y}_i^{(\text{clu})}$ according to this partition.

Based on the clustering result and classifier prediction, the refined pseudolabel is determined by jointly considering clustering consistency and classifier confidence. These two criteria provide complementary measures of pseudolabel reliability. Clustering consistency captures the structural organization of target-domain features, whereas classifier confidence quantifies prediction certainty. Under substantial cross-scene domain shifts, however, the classifier produces overconfident erroneous predictions. Conversely, clustering consistency becomes unreliable near category boundaries or for categories with similar spectral signatures. To reduce the risk of accepting erroneous pseudolabels based on either criterion alone, SSIM combines the two criteria during pseudolabel selection. In particular, the predicted label of a target-domain sample is accepted as a known-class pseudolabel only when its cluster assignment agrees with the classifier prediction and the corresponding prediction confidence exceeds the class-adaptive threshold. Among the samples that do not satisfy these conditions, those belonging to $\mathcal{U}$ are assigned to the unknown category, while the remaining samples are left unlabeled, i.e.,

$$\tilde{y}_i = \begin{cases} \hat{y}_i, & \text{if } \hat{y}_i^{(\text{clu})} = \hat{y}_i \text{ and } p_i^{(\hat{y}_i)} > \tau_{\hat{y}_i} \\ C+1, & \text{if } \mathbf{x}_i \in \mathcal{U} \\ -1, & \text{otherwise} \end{cases} \tag{10}$$

where -1 indicates that $\mathbf{x}_i$ is not assigned any pseudolabel. Accordingly, samples that do not satisfy the pseudolabel selection criteria remain unlabeled at this stage and do not contribute to the pseudolabel-based target-domain losses, reducing the risk of propagating erroneous supervisory signals.

Using these refined pseudolabels, target-domain samples designated as known or unknown are selected for self-supervised optimization. However, high-confidence screening

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can cause severe category imbalance among the selected known categories. To mitigate this issue, we further introduce a categorywise completion strategy. For each underrepresented known category, additional target-domain samples that have the highest remaining confidence and match the predicted category are selected to supplement the pseudolabeled set. The minimum completion ratio for each known category is governed by a hyperparameter ρ. This constrained categorywise completion mitigates category imbalance among the selected pseudolabels while maintaining their reliability, providing more stable pseudolabel supervision for target-domain adaptation. After completion, the final sets of target-domain samples with known and unknown pseudolabels are denoted by $\mathcal{T}^k$ and $\mathcal{T}^u$, respectively.

The selected target domain samples with known pseudolabels provide self-supervision for target-domain adaptation. The corresponding pseudolabel classification loss is formulated as

$$\mathcal{L}_{\text{pk_cls}} = -\frac{1}{|\mathcal{T}^k|}\sum_{\mathbf{x}_i\in\mathcal{T}^k}\sum_{c=1}^{C}\mathbb{I}(\tilde{y}_i = c)\log p_i^{(c)} \tag{11}$$

where $\mathbb{I}(\cdot)$ is the indicator function. To further enhance intraclass compactness and feature consistency, we introduce a prototype consistency loss

$$\mathcal{L}_{\text{pk_proto}} = \frac{1}{|\mathcal{T}^k|}\sum_{\mathbf{x}_i\in\mathcal{T}^k}\left\|\mathbf{f}_i - \boldsymbol{\mu}_{\tilde{y}_i}\right\|_2^2 \tag{12}$$

where $\boldsymbol{\mu}_{\tilde{y}_i}$ denotes the prototype center of category $\tilde{y}_i$.

For target domain samples with unknown pseudolabels, directly pulling them toward known category centers would blur the boundary between known and unknown categories. Therefore, we introduce a repulsion loss

$$\mathcal{L}_{\text{pu_rep}} = \frac{1}{|\mathcal{T}^u|}\sum_{\mathbf{x}_i\in\mathcal{T}^u}\max\left(0,\, m - \min_{c\in\{1,\ldots,C\}}\left\|\mathbf{f}_i - \boldsymbol{\mu}_c\right\|_2\right) \tag{13}$$

where m is a margin parameter that controls the minimum desired distance between unknown-candidate samples and known category centers. This loss also establishes a more reliable foundation for subsequent pseudolabel assignment in the target domain.

The supervised classification loss on the labeled source domain is defined as

$$\mathcal{L}_{\text{src}} = -\frac{1}{|\mathcal{S}|}\sum_{(\mathbf{x}_i,y_i)\in\mathcal{S}}\sum_{c=1}^{C}\mathbb{I}(y_i = c)\log p_i^{(c)}. \tag{14}$$

The overall optimization objective of CF-CAN is formulated as

$$\mathcal{L} = \mathcal{L}_{\text{src}} + \mathcal{L}_{\text{pk_cls}} + \lambda\left(\mathcal{L}_{\text{pk_proto}} + \mathcal{L}_{\text{pu_rep}}\right) \tag{15}$$

where λ is a balancing coefficient. The prototype consistency loss encourages pseudoknown samples to align with their corresponding category prototypes, whereas the repulsion loss pushes pseudounknown samples away from known-class prototypes. Combined with the selective use of screened target samples, these complementary constraints help limit the influence of residual pseudolabel errors during model optimization. Through this design, the proposed SSIM module provides more reliable supervision from unlabeled target-domain data. Consequently, it facilitates cross-scene adaptation and improves the separability between known and unknown categories.

IV. Experiment

A. Datasets

To evaluate the effectiveness of the proposed method, experiments are conducted on six hyperspectral datasets: Pavia University, Pavia Center, Houston2013, Houston2018, YRD_GF5_11 and YRD_GF5_5. From these datasets, three open-set cross-scene HSIC tasks are constructed, namely PaviaU-C, Houston13–18, and YRD11-5. In each task, the former and latter datasets in the pair are treated as the source domain and target domain, respectively. The open-set setting is established by selecting a subset of shared classes from both domains as the known categories. In addition, one category present in the target domain but absent from the source domain label space is designated as the unknown category.

The PaviaU-C task is constructed from the Pavia University and Pavia Center datasets. Both were acquired by the ROSIS sensor over Pavia, northern Italy. Their spatial sizes are 610 × 340 and 1096 × 715 pixels, respectively. After preprocessing, the two datasets retain 103 and 102 spectral bands. The 102 common bands are used in the experiments, and six shared categories are treated as the known categories.

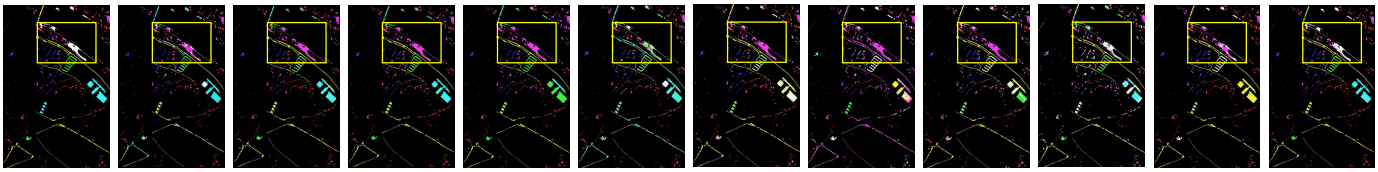
The Houston13–18 task consists of two datasets acquired over Houston, USA: Houston2013 and Houston2018. Houston2013 has a spatial size of 349 × 1905 pixels with 144 spectral bands. Houston2018 has a spatial size of 601 × 2384 pixels with 48 spectral bands. For this task, the 48 common spectral bands shared by the two datasets are used, and six shared categories are selected as the known categories.

The YRD11-5 task involves the YRD_GF5_11 and YRD_GF5_5 datasets. They were acquired by the AHSI sensor onboard China's GaoFen-5 satellite over the Yellow River Delta. YRD_GF5_11 was collected on November 5, 2022, and YRD_GF5_5 was collected on May 25, 2022. Their spatial sizes are 729 × 1076 and 2009 × 2104 pixels, respectively. After removing low-SNR and water-vapor absorption bands, both datasets retain 150 effective spectral channels. In the experiments, nine shared categories between the two datasets are used as the known categories.

B. Experimental Setting

The proposed CF-CAN method is implemented using the PyTorch framework and trained on a single NVIDIA GeForce RTX 3090 GPU. All experiments follow a labeled source, unlabeled target OSDA protocol, where labeled source samples and unlabeled target samples are used during training. Target domain labels are withheld throughout training and used solely for evaluation. In particular, the proposed SSIM module leverages the unlabeled target samples to reliably mine pseudoknown and pseudounknown instances, facilitating domain adaptation. In the source domain, 80% of the samples are used for training and the remaining 20% for validation. The input hyperspectral image patches are set to a spatial size of 13 × 13. Training employs the Adam optimizer with a fixed

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 2. Visualization of the classification maps of various HSIC methods on the PaviaU-C task. (a) Ground truth image. (b) OSC-SCL. (c) D3Net. (d) FDGNet. (e) ASAL. (f) FRFSL. (g) WGDT. (h) PWDA. (i) JDMUS. (j) OSUSE. (k) SoDa^2. (l) CF-CAN.

TABLE I

EXPERIMENT COMPARISON WITH THE STATE-OF-THE-ART HSIC METHODS ON THE PAVIAU-C TASK

Category	OSC-SCL [27]	D3Net [28]	FDGNet [29]	ASAL [30]	FRFSL [31]	WGDT [22]	PWDA [23]	JDMUS [24]	OSUSE [25]	SoDa^2 [26]	CF-CAN (ours)
1	88.40	91.72	80.18	59.50	83.91	85.40	34.00	65.53	26.06	**95.80**	92.66
2	90.36	90.45	84.25	88.87	56.23	**96.79**	43.79	95.90	87.74	95.84	93.47
3	65.07	93.71	94.67	94.64	87.54	34.64	72.55	30.17	73.22	6.74	**95.68**
4	68.88	84.89	86.15	68.86	68.17	0.69	0.43	1.92	48.62	1.19	**86.19**
5	91.57	97.73	83.93	89.63	96.71	94.62	87.11	89.66	26.86	99.69	**99.83**
6	48.89	74.53	76.25	87.15	89.98	95.15	**99.48**	95.73	94.47	91.39	90.91
Unknown	39.78	1.84	1.18	1.40	57.90	18.10	27.29	21.52	57.91	42.00	**94.41**
Closed OA	73.95±0.02	83.90±2.51	85.75±0.74	69.86±5.56	77.03±3.39	62.62±0.02	34.47±9.13	62.09±0.03	43.04±0.11	71.30±0.02	**90.90±1.11**
Closed AA	79.81±0.03	82.20±4.21	86.03±1.03	76.33±4.53	80.19±2.91	61.46±0.03	43.93±11.69	61.17±0.03	49.47±0.10	71.96±0.03	**91.03±1.43**
Closed Kappa	68.33±0.02	79.66±0.03	82.16±0.01	63.29±6.76	71.89±0.04	55.67±0.02	26.10±10.13	55.10±0.03	36.02±0.12	63.21±0.02	**88.55±0.01**
Open OA	64.64±0.05	69.51±2.15	71.28±0.79	58.27±4.63	72.40±7.37	55.00±0.02	30.25±5.55	54.50±0.02	41.63±0.10	48.08±0.14	**90.42±1.06**
Open AA	66.59±0.04	70.07±3.59	73.58±0.85	65.50±3.88	75.41±6.12	55.12±0.03	38.97±8.31	54.82±0.02	47.35±0.09	50.83±0.12	**90.36±1.44**
Open Kappa	57.99±0.05	63.80±0.03	66.09±0.01	51.49±5.37	67.23±0.08	46.86±0.02	20.90±6.60	46.37±0.03	31.80±0.11	37.80±0.16	**88.48±0.01**

learning rate of $1e-3$ and a maximum of 400 epochs. The threshold scaling factor γ, the minimum completion ratio ρ, and the size R of the initial unknown candidate pool are set to 0.8%, 10%, and 3000, respectively. During evaluation, we adopt several widely used HSIC metrics: per-class accuracy, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. All experiments are repeated ten times, and the mean values with standard deviations are reported to ensure a fair comparison.

C. Comparison With State-of-the-Art HSIC Methods

To validate the effectiveness of CF-CAN, we compare it with some state-of-the-art HSIC methods, including OSC-SCL [27], D3Net [28], FDGNet [29], ASAL [30], FRFSL [31], WGDT [22], PWDA [23], JDMUS [24], OSUSE [25], and SoDa^2 [26]. These methods fall into four paradigms: open-set HSIC, domain generalization, cross-domain few-shot learning, and OSDA. Non-OSDA methods are adapted to our protocol with labeled source data and unlabeled target data and used as supplementary baselines. Those that follow the same OSDA protocol as CF-CAN serve as primary baselines. Both groups are reported, but the main conclusions are based primarily on comparisons with the primary baselines. The supplementary baselines are included for reference only.

As shown in Table I, CF-CAN achieves the best performance for most known categories and the unknown category on the PaviaU-C task. It attains an unknown accuracy of 94.41% and an Open OA of 90.42%, both significantly outperforming these competing methods. CF-CAN also yields the best performance on the remaining closed-set and open-set metrics. Several representative cross-scene HSIC methods, such as D3Net and FDGNet, achieve relatively competitive performance in terms of Closed OA, reaching 83.90% and 85.75%, respectively. However, their unknown accuracies drop sharply to 1.84% and 1.18%. The open-set HSIC method OSC-SCL achieves a higher unknown accuracy of 39.78%, yet its Closed OA and Open OA remain at 73.95% and 64.64%, respectively. Recent OSDA methods achieve competitive performance on several known categories. In particular, OSUSE and SoDa^2 attain unknown-class accuracies of 57.91% and 42.00%, and outperform WGDT, PWDA, and JDMUS. Despite these improvements, both methods still underperform CF-CAN in terms of Open OA and unknown-class accuracy. These results indicate that CF-CAN can simultaneously achieve accurate recognition of known categories and reliable detection of unknown categories under cross-scene conditions. The visual results in Fig. 2 further support this conclusion. Several comparison methods still exhibit fragmented regions and scattered misclassified pixels, reflecting weaker spatial consistency in complex areas. On the contrary, CF-CAN produces more spatially coherent classification maps with more complete regions and clearer category boundaries.

On the Houston13–18 task, CF-CAN again achieves superior performance. As reported in Table II, it obtains the highest Closed OA of 92.99% and Open OA of 85.89%. It also ranks first on the other overall evaluation metrics. Among the recent OSDA methods, SoDa^2 attains a relatively high unknown-class accuracy of 56.67%. PWDA performs even better on this metric, reaching 60.12%, which is close to the 61.81% achieved by CF-CAN. However, the Open OA of PWDA is only 56.97%. On the contrary, CF-CAN attains the highest values for both Open OA and unknown-class accuracy. These results suggest that strong unknown-class detection alone does not necessarily guarantee superior overall performance in open-set cross-scene HSIC. CF-CAN offers a better balance between accurate known-class recognition and reliable unknown-class detection. The visual comparisons in Fig. 3 further support this interpretation. In complex urban scenes, several compared methods

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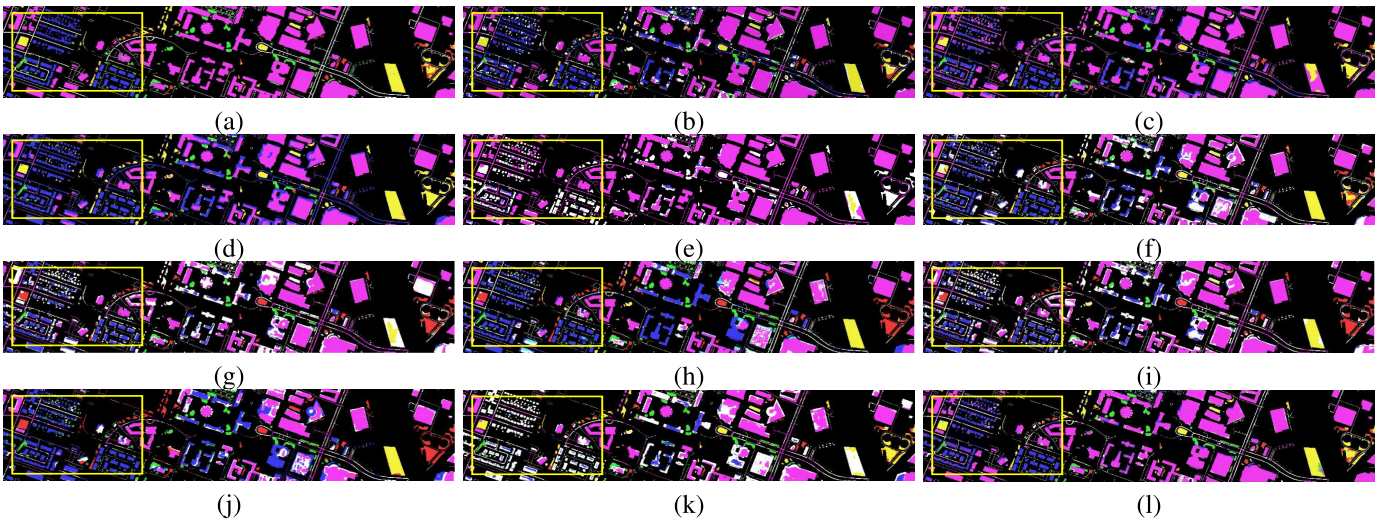


Fig. 3. Visualization of the classification maps of various HSIC methods on the Houston13–18 task. (a) Ground truth image. (b) OSC-SCL. (c) D3Net. (d) FDGNet. (e) ASAL. (f) FRFSL. (g) WGDT. (h) PWDA. (i) JDMUS. (j) OSUSE. (k) SoDa^2. (l) CF-CAN.

TABLE II

EXPERIMENT COMPARISON WITH THE STATE-OF-THE-ART HSIC METHODS ON THE HOUSTON13–18 TASK

Category	OSC-SCL [27]	D3Net [28]	FDGNet [29]	ASAL [30]	FRFSL [31]	WGDT [22]	PWDA [23]	JDMUS [24]	OSUSE [25]	SoDa^2 [26]	CF-CAN (ours)
1	72.40	65.85	23.06	72.36	74.55	1.85	**97.56**	93.79	93.87	53.81	76.94
2	82.33	85.68	75.55	52.41	61.05	78.15	43.27	32.79	34.29	56.42	**93.39**
3	68.02	64.68	55.06	20.21	48.06	56.22	68.51	29.65	63.09	68.80	**70.64**
4	94.12	**100.00**	**100.00**	45.45	94.12	95.45	**100.00**	59.09	27.27	45.45	**100.00**
5	78.34	77.95	82.19	67.05	82.87	13.11	**83.52**	58.00	82.48	11.71	82.91
6	51.24	93.54	85.10	83.02	55.33	65.35	65.72	60.74	47.70	53.39	**96.11**
Unknown	46.35	2.67	1.14	16.75	50.29	55.65	60.12	51.50	42.09	56.67	**61.81**
Closed OA	54.88±0.06	80.21±1.70	78.47±2.66	75.73±2.88	61.15±6.68	53.21±0.08	55.31±0.06	52.85±0.09	43.32±0.09	72.19±0.07	**92.99±0.95**
Closed AA	77.28±0.01	65.03±4.12	76.76±3.38	52.90±6.60	65.96±8.84	51.14±0.04	71.43±0.04	51.27±0.04	48.28±0.13	73.92±0.04	**86.73±1.33**
Closed Kappa	37.89±0.03	57.90±0.04	61.32±0.04	54.46±4.30	42.10±0.06	33.44±0.05	37.31±0.05	33.33±0.06	27.54±0.08	55.10±0.07	**85.70±0.02**
Open OA	49.60±0.03	70.25±1.52	67.97±2.21	68.01±2.29	59.17±4.65	54.31±0.05	56.97±0.05	53.98±0.06	42.96±0.08	54.93±0.08	**85.89±1.81**
Open AA	65.59±0.02	55.33±3.29	64.35±2.85	46.95±5.63	63.66±6.09	52.75±0.04	71.11±0.03	52.85±0.04	47.14±0.11	57.52±0.10	**80.95±1.68**
Open Kappa	34.97±0.02	44.33±0.04	48.00±0.03	42.79±3.48	42.60±0.05	33.97±0.04	42.12±0.05	33.76±0.05	27.36±0.07	36.81±0.08	**75.90±0.03**

TABLE III

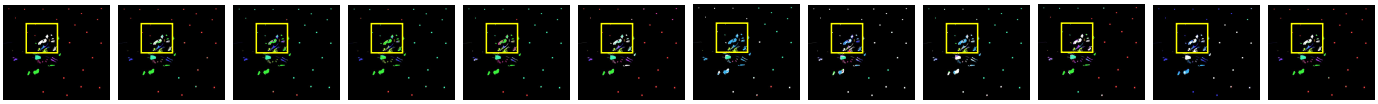
EXPERIMENT COMPARISON WITH THE STATE-OF-THE-ART HSIC METHODS ON THE YRD11-5 TASK

Category	OSC-SCL [27]	D3Net [28]	FDGNet [29]	ASAL [30]	FRFSL [31]	WGDT [22]	PWDA [23]	JDMUS [24]	OSUSE [25]	SoDa^2 [26]	CF-CAN (ours)
1	57.07	55.10	20.81	17.95	82.66	8.54	1.47	1.17	70.98	2.16	**99.88**
2	95.87	96.48	61.73	94.13	95.57	4.47	75.51	11.88	**100.00**	55.28	97.21
3	99.37	99.81	99.98	**100.00**	93.54	99.78	5.20	99.97	87.91	3.33	**100.00**
4	97.39	99.58	99.76	78.05	64.80	0.10	17.54	6.08	88.16	0.01	**99.86**
5	88.97	89.74	27.10	95.04	92.99	12.32	88.13	0.36	77.09	5.41	**95.70**
6	31.60	1.33	0.00	66.25	35.04	22.93	11.42	36.48	0.69	41.93	**83.43**
7	72.09	84.60	87.25	89.81	65.66	85.06	35.38	49.45	0.43	**94.05**	91.43
8	33.28	1.47	0.52	16.76	80.89	0.08	4.16	0.14	69.16	2.85	**86.39**
9	55.92	13.62	3.96	86.47	74.80	67.62	89.23	58.51	40.68	3.15	**95.19**
Unknown	56.29	7.79	0.00	15.87	69.57	27.32	26.74	29.93	80.89	61.13	**93.89**
Closed OA	59.76±0.29	66.42±5.08	59.40±3.22	58.26±4.95	75.98±1.94	20.78±0.05	21.50±0.02	15.83±0.02	63.59±0.04	36.62±0.06	**94.84±0.77**
Closed AA	54.52±0.35	54.20±4.59	51.01±3.60	50.35±5.49	74.96±3.02	25.76±0.06	29.43±0.01	21.43±0.01	55.05±0.03	42.31±0.02	**91.09±1.95**
Closed Kappa	50.86±0.36	61.00±0.06	50.50±0.04	50.71±5.52	72.23±0.02	15.61±0.05	17.42±0.02	11.42±0.01	57.89±0.04	30.06±0.06	**93.74±0.01**
Open OA	59.09±0.29	50.07±2.25	45.00±2.44	46.53±3.78	68.42±8.96	18.62±0.02	20.83±0.01	14.81±0.04	62.49±0.10	26.63±0.04	**93.00±1.07**
Open AA	56.33±0.34	48.27±4.21	45.91±3.24	46.31±5.05	72.59±4.04	24.38±0.05	28.36±0.01	20.45±0.03	55.45±0.05	27.17±0.01	**90.06±1.98**
Open Kappa	50.51±0.33	43.10±0.04	35.89±0.03	37.91±4.10	65.13±0.09	9.36±0.03	7.15±0.01	3.91±0.03	55.88±0.11	15.43±0.04	**91.68±0.01**

tend to generate discontinuous category regions and confused spatial structures. This reflects insufficient spatial consistency and weaker discrimination between neighboring categories. Compared with these methods, CF-CAN yields more complete category regions, stronger local spatial consistency, and less confusion between adjacent categories. Its classification maps better match the spatial organization of the target scene.

Table III shows that CF-CAN maintains superior performance on the YRD11-5 task. CF-CAN achieves an unknown accuracy of 93.89% and an Open OA of 93.00%, both markedly higher than those of the competing methods. It also obtains the best performance on the other closed-set and open-set metrics, further demonstrating its comprehensive classification capability. Compared with PaviaU-C and Houston13–18, the YRD11-5 task is more challenging. This is because the target scene is relatively sparse and the spatial distribution of target categories is less dense, making the distinction between known and unknown areas more difficult. Under this challenging setting, the limitations of the competing methods become more evident. For example, FRFSL achieves the highest Open OA among the competing methods at 68.42%, but its unknown accuracy is 69.57%.

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 4. Visualization of the classification maps of various HSIC methods on the YRD11-5 task. (a) Ground truth image. (b) OSC-SCL. (c) D3Net. (d) FDGNet. (e) ASAL. (f) FRFSL. (g) WGDT. (h) PWDA. (i) JDMUS. (j) OSUSE. (k) SoDa^2. (l) CF-CAN.

TABLE IV
ABLATION EXPERIMENTS ON THE PAVIAU-C AND HOUSTON13–18 TASKS

Dataset	AMFF	CADT	SSIM	Closed OA	Closed AA	Closed Kappa	Open OA	Open AA	Open Kappa
PaviaU-C	-	-	-	82.91±1.50	84.29±1.36	78.67±0.02	58.75±1.98	62.38±1.55	51.36±0.02
	✓	-	-	84.34±1.04	85.28±0.79	80.43±0.01	60.88±1.90	63.58±1.09	53.81±0.02
	-	✓	-	84.52±0.89	85.52±0.55	80.68±0.01	63.28±0.74	65.10±0.47	56.54±0.01
	✓	-	✓	88.51±0.26	88.59±0.66	85.64±0.00	79.49±1.56	76.74±2.25	75.25±0.02
	-	✓	✓	89.33±0.77	89.18±0.09	86.72±0.01	89.50±0.84	89.81±0.58	87.41±0.01
	✓	✓	-	85.65±1.04	86.29±0.79	82.06±0.01	70.23±0.86	73.10±0.68	64.92±0.01
	✓	✓	✓	**90.90±1.11**	**91.03±1.43**	**88.55±0.01**	**90.42±1.06**	**90.36±1.44**	**88.48±0.01**
Houston13–18	-	-	-	76.68±2.15	46.39±6.76	38.06±0.06	65.78±2.75	43.65±3.59	32.92±0.04
	✓	-	-	78.66±0.62	59.28±5.80	53.11±0.03	66.48±1.06	46.43±7.67	38.67±0.04
	-	✓	-	78.78±1.10	59.65±4.87	54.82±0.03	68.33±0.97	46.85±4.17	38.88±0.04
	✓	-	✓	87.89±0.34	85.28±0.86	83.13±0.01	74.86±1.19	71.23±0.92	69.24±0.01
	-	✓	✓	91.47±1.09	85.48±0.43	85.11±0.02	84.67±1.86	77.42±1.34	73.12±0.03
	✓	✓	-	82.01±2.02	60.43±8.66	56.64±0.08	69.80±0.54	53.16±4.97	42.83±0.04
	✓	✓	✓	**92.99±0.95**	**86.73±1.33**	**85.70±0.02**	**85.89±1.81**	**80.95±1.68**	**75.90±0.03**

Meanwhile, OSUSE obtains the highest unknown accuracy among the competing methods at 80.89%, while its Open OA is 62.49%. Nevertheless, both methods remain clearly below CF-CAN in terms of Open OA and unknown accuracy. The sparse target distribution and ambiguous boundaries between known and unknown categories pose considerable challenges to simultaneously achieving effective cross-scene adaptation and reliable unknown category discrimination. On the contrary, CF-CAN emphasizes collaborative feature and classifier adaptation. It jointly strengthens robust feature representation, calibrates class-adaptive rejection boundaries, and ensures reliable mining of both pseudoknown and pseudounknown target samples. This collaborative design enables CF-CAN to effectively handle sparse target distributions and ambiguous boundaries between known and unknown regions, yielding more stable and robust open-set cross-scene performance. The visual results in Fig. 4 further confirm this conclusion. The competing methods tend to produce more scattered misclassified pixels in background regions and exhibit more noticeable confusion between known and unknown areas. By contrast, CF-CAN generates cleaner and more concentrated classification results, better preserves the spatial distribution of target categories, and displays fewer isolated errors in background regions.

Overall, the above results consistently demonstrate that existing methods perform well in either cross-scene adaptation or open-set recognition, but seldom in both. Cross-scene HSIC methods generally perform well on seen categories, yet they often lack reliable rejection of unknown categories. Open-set HSIC methods are more effective at identifying unknown categories, but they are usually more vulnerable to cross-scene distribution shifts. While recent OSDA HSIC methods account for domain shifts and unknown categories, they fall short in complex cross-scene scenarios, where they cannot both accurately identify known categories and dependably reject unknowns. By contrast, CF-CAN consistently outperforms them across all three tasks. This verifies the effectiveness of the proposed framework. It jointly enhances recognition of known categories, rejection of unknown categories, and cross-scene generalization, providing a more effective solution for open-set cross-scene HSIC.

D. Ablation Studies

We analyze the contribution of the AMFF module, the CADT strategy, and the SSIM module in CF-CAN through ablation studies on the PaviaU-C and Houston13–18 tasks. The results are reported in Table IV.

For the PaviaU-C task, the baseline without the AMFF module, the CADT strategy, or the SSIM module yields a Closed OA of 82.91% and an Open OA of 58.75%. Introducing either the AMFF module or the CADT strategy alone brings only limited performance improvements. The CADT strategy yields slightly better performance than the AMFF module, which suggests that classifier-level boundary calibration is more directly related to category discrimination under domain shift. Notably, the variant without CADT retains threshold-based rejection using a carefully tuned global threshold. Relative to this baseline, introducing CADT increases Closed OA from 82.91% to 84.52% and Open OA from 58.75% to 63.28%. These improvements of 1.61 and 4.53 percentage points demonstrate the advantage of CADT over a single global threshold. When the AMFF module and the CADT strategy are combined, the Closed OA improves to 85.65% and the Open OA to 70.23%. However, the overall gain remains limited. This indicates that feature adaptation and threshold

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TABLE V

ABLATION STUDY OF DIFFERENT FUSION STRATEGIES IN THE AMFF MODULE ON THE HOUSTON13–18 TASK

Metric	Concatenation	Addition	Unidirectional-T	Unidirectional-C	Bidirectional
Closed OA	83.67±2.87	85.15±4.76	82.05±2.76	87.62±1.39	**92.99±0.95**
Closed AA	70.32±2.16	74.14±7.20	59.14±5.54	72.08±7.72	**86.73±1.33**
Closed Kappa	67.27±4.78	70.13±9.15	62.71±5.45	74.21±3.39	**85.70±0.02**
Open OA	73.66±2.52	75.21±4.45	72.23±2.43	77.99±2.08	**85.89±1.81**
Open AA	60.28±1.85	64.03±6.66	50.69±4.75	63.41±6.88	**80.95±1.68**
Open Kappa	52.64±3.05	55.58±7.25	48.51±4.12	59.49±4.73	**75.90±0.03**

adjustment alone are insufficient for the open-set cross-scene setting. On the contrary, introducing the SSIM module results in substantially greater performance improvements. In particular, the combination of the AMFF module and the SSIM module yields a Closed OA of 88.51% and an Open OA of 79.49%. The combination of the CADT strategy and the SSIM module achieves 89.33% and 89.50% on the corresponding metrics. These results suggest that the SSIM module is crucial for exploiting reliable target-domain information, thereby facilitating category alignment and discriminative separation among target samples. They also indicate that the CADT strategy becomes more effective at calibrating class-adaptive decision boundaries when reliable target-domain information is available.

For the Houston13–18 task, a similar trend is observed, and the performance gains are more pronounced. The baseline yields a Closed OA of 76.68% and an Open OA of 65.78%. Introducing the AMFF module alone improves the Closed OA to 78.66% and the Open OA to 66.48%. When applied independently, CADT outperforms the baseline with a carefully tuned global threshold, achieving Closed OA and Open OA values of 78.78% and 68.33%, respectively. Therefore, the CADT strategy proves more effective than the AMFF module when used individually, especially for open-set recognition. On the contrary, the introduction of the SSIM module leads to a substantial jump in performance. The combination of the AMFF module and the SSIM module improves the Closed OA to 87.89% and the Open OA to 74.86%. When the CADT strategy is combined with the SSIM module, these metrics increase further to 91.47% and 84.67%, respectively. These larger gains highlight the critical importance of reliable target-domain information for open-set cross-scene HSIC. This confirms that the SSIM module acts as a key bridge between feature adaptation and classifier adaptation.

To assess the effectiveness of AMFF, we conduct additional experiments on the Houston13–18 task from two perspectives: feature fusion performance and source–target feature discrepancy. We first compare five fusion strategies: concatenation, addition, unidirectional-T, unidirectional-C, and bidirectional. All variants use the same feature extraction branches and training settings, differing only in their fusion mechanisms. Unidirectional-T allows information to flow only from the CNN branch to the dynamic Window-Transformer branch, whereas unidirectional-C allows information to flow only in the opposite direction. The bidirectional variant enables reciprocal information exchange between the two branches. As shown in Table V, concatenation, addition, unidirectional-T, and unidirectional-C achieve Open OA values of 73.66%, 75.21%, 72.23%, and 77.99%, respectively. The bidirectional

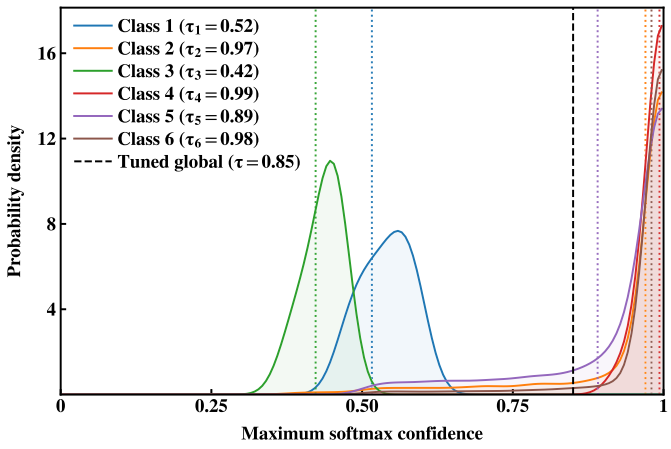


Fig. 5. Classwise distributions of maximum softmax confidence with class-adaptive thresholds and the tuned global threshold on the Houston13–18 task.

variant achieves the highest Open OA of 85.89%, outperforming the best-performing alternative, unidirectional-C, by 7.90 percentage points. It also achieves the best results across all other reported closed-set and open-set metrics. These results indicate that direct feature fusion and unidirectional interaction are less effective at exploiting the complementary information provided by the two branches. By enabling reciprocal refinement of branch-specific representations, bidirectional cross-attention more effectively integrates the local spectral–spatial patterns captured by the CNN branch with the contextual representations learned by the dynamic Window-Transformer branch. We also perform a posthoc maximum mean discrepancy (MMD) analysis to quantify the contribution of AMFF to reducing the empirical source–target feature discrepancy. All other components and experimental settings are kept unchanged. The MMD value decreases from 0.15 ± 0.06 without AMFF to 0.05 ± 0.02 with AMFF, corresponding to a relative reduction of 66.67%. This result indicates that the complete framework with AMFF produces feature representations with a smaller empirical discrepancy between the source and target distributions. Although this reduction does not establish strict domain invariance, it provides evidence that AMFF facilitates the learning of more transferable representations within the overall adaptation framework.

To provide an intuitive analysis of CADT, we visualize the classwise distributions of maximum softmax confidence scores on the Houston13–18 task. The density curves show how target samples are distributed across different confidence levels, with the corresponding class-adaptive thresholds and the tuned global threshold overlaid for comparison. As shown in Fig. 5, the confidence distributions vary considerably across categories in the cross-scene setting. Some categories exhibit relatively low confidence levels, whereas others are concentrated at substantially higher confidence levels. Consequently, a single global threshold, even when carefully tuned, is overly restrictive for some categories and overly permissive for others. CADT addresses this limitation by adapting the threshold to the class-specific confidence distribution, enabling more reliable discrimination between known and unknown target samples.

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TABLE VI
ABLATION STUDY OF PSEUDOLABEL SELECTION CRITERIA IN THE SSIM MODULE ON THE HOUSTON13–18 TASK

Metric	Clustering-only	Confidence-only	Combined
Pseudo-label Accuracy	69.07±2.01	72.61±4.20	**82.10±3.84**
Open OA	69.31±0.38	70.02±0.88	**85.89±1.81**
Open AA	41.46±2.41	45.17±8.77	**80.95±1.68**
Open Kappa	36.38±2.06	42.42±5.43	**75.90±0.03**

We further examine the complementary roles of clustering consistency and classifier confidence in SSIM through an additional ablation study on the Houston13–18 task. In particular, the clustering-only variant uses the clustering consistency criterion without imposing the classifier confidence constraint, whereas the confidence-only variant uses the classifier confidence criterion without imposing the clustering consistency constraint. The Combined variant jointly applies both criteria. Pseudolabel accuracy is defined as the proportion of correctly assigned pseudolabels among the target samples selected for pseudolabeling. The ground-truth target labels are used solely to compute this metric and are not involved in training or pseudolabel selection. As shown in Table VI, the clustering-only and confidence-only variants achieve pseudolabel accuracies of 69.07% and 72.61%, respectively. The combined variant increases the accuracy to 82.10%, corresponding to gains of 13.03 and 9.49 percentage points over the two individual variants, respectively. It also yields substantial improvements in Open OA, Open AA, and Open Kappa. These results demonstrate that jointly leveraging the structure of target-domain features and classifier confidence produces more reliable pseudolabel supervision and improves downstream classification performance.

When all three components are jointly enabled, CF-CAN achieves the best performance on both tasks. More importantly, the full model consistently outperforms the combination of the CADT strategy and the SSIM module. This two-module configuration delivers the best performance among all setups that use only two components. Compared with that configuration, the full model improves the Closed OA from 89.33% to 90.90% and the Open OA from 89.50% to 90.42% for the PaviaU-C task. For the Houston13–18 task, the Closed OA increases from 91.47% to 92.99%, while the Open OA improves from 84.67% to 85.89%. This demonstrates that the AMFF module remains valuable even after reliable target-domain supervision and class-adaptive threshold calibration have been established. In other words, once the SSIM module provides informative target cues and the CADT strategy calibrates the decision boundaries, the AMFF module can further learn more transferable and discriminative multiscale representations. This allows the entire framework to operate on a more stable feature space. Overall, these results demonstrate that CF-CAN functions as a collaborative framework. In this framework, the AMFF module mainly enhances representation robustness under domain shift, and the CADT strategy improves classifier-level boundary calibration for more reliable open-set recognition. Meanwhile, the SSIM module supplies reliable target-domain supervision to bridge feature adaptation and classifier adaptation. Their collaboration is essential for robust open-set cross-scene HSIC.

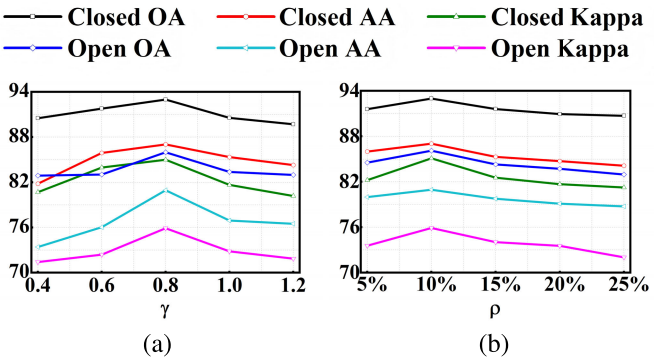


Fig. 6. Sensitivity analysis of CF-CAN on the Houston13–18 task. (a) Threshold scaling factor γ. (b) Minimum completion ratio ρ.

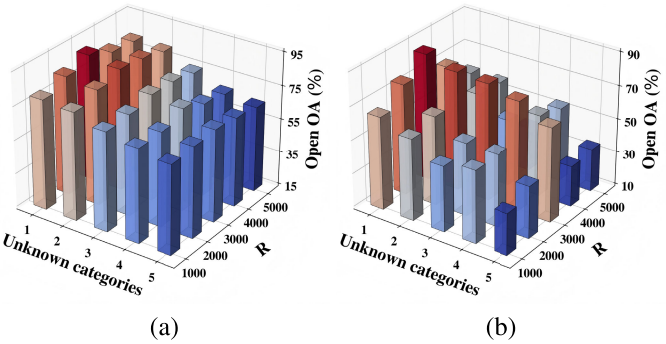


Fig. 7. Sensitivity analysis of R on the PaviaU-C and Houston13–18 tasks with different openness levels. (a) PaviaU-C task. (b) Houston13–18 task.

E. Parameters Sensitivity

To evaluate the robustness of CF-CAN, we analyze the model sensitivity of two key hyperparameters on the Houston13–18 task. As shown in Fig. 6(a), both closed-set and open-set metrics initially improve and then decline as the threshold scaling factor γ increases from 0.4 to 1.2. The best overall performance is achieved at $\gamma = 0.8$. Compared with $\gamma = 0.4$, setting $\gamma = 0.8$ yields approximate gains of 2.48% in Closed OA and 3.00% in Open OA. It also outperforms the $\gamma = 1.2$ setting by roughly 3.28% in Closed OA and 2.91% in Open OA. These findings suggest that an appropriate threshold adjustment helps CF-CAN accurately recognize known categories and reliably detect unknown ones under cross-scene conditions. When γ is too small, the rejection criterion becomes overly strict. Consequently, some informative samples in the target domain cannot be effectively leveraged to facilitate the adaptation of the source domain model to the target domain. On the contrary, an excessively large γ loosens the rejection boundary too much. This introduces unreliable pseudolabeled samples and weakens the discrimination between known and unknown categories.

A similar trend is observed for the minimum completion ratio ρ in Fig. 6(b). The overall performance peaks at $\rho = 10\%$. In particular, compared with $\rho = 5\%$, setting $\rho = 10\%$ improves Closed OA and Open OA by approximately 1.40% and 1.36%, respectively. When ρ is further increased to 25%, the performance drops noticeably. Relative to the best setting ($\rho = 10\%$), Closed OA decreases by

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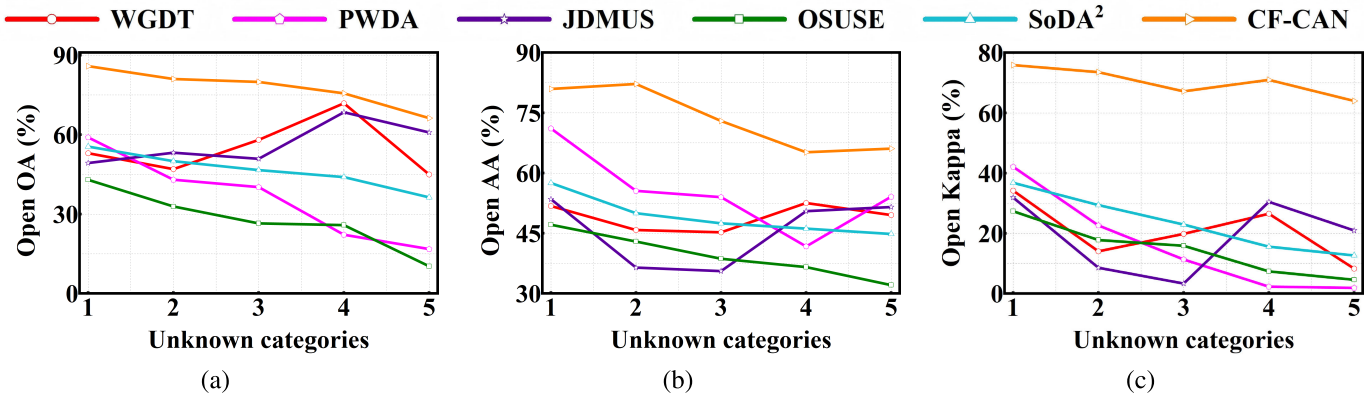


Fig. 8. Experiment comparison with the state-of-the-art HSIC methods on the Houston13–18 task with different unknown categories. (a) Open OA. (b) Open AA. (c) Open Kappa.

about 2.28% and Open OA by 2.93%. This indicates that an appropriate minimum completion ratio facilitates the utilization of reliable target-domain information during adaptation. However, an excessively large completion ratio introduces more noisy pseudolabeled samples, impairing both feature adaptation and boundary calibration. Overall, these results demonstrate that CF-CAN benefits from properly calibrated rejection boundaries and a suitable amount of reliable target-domain supervision. We adopt $\gamma = 0.8$ and $\rho = 10\%$ in all subsequent experiments.

To examine the influence of R across different openness settings, we conduct a joint sensitivity analysis on the PaviaU-C and Houston13–18 tasks. In particular, R is varied over $\{1000, 2000, 3000, 4000, 5000\}$, while the number of unknown categories ranges from one to five. As shown in Fig. 7, when $R = 3000$, increasing the number of unknown categories from one to five reduces the Open OA by 20.28 percentage points on PaviaU-C, from 90.42% to 70.14%, and by 19.63 percentage points on Houston13–18, from 85.89% to 66.26%. This overall downward trend reflects the increasing difficulty of classification under higher openness settings. When the number of unknown categories is fixed at one, $R = 3000$ achieves Open OA values of 90.42% and 85.89% on PaviaU-C and Houston13–18, respectively. Compared with $R = 1000$, these results represent improvements of 9.86 and 20.45 percentage points, respectively; compared with $R = 5000$, the corresponding improvements are 7.34 and 27.31 percentage points. Across all evaluated openness settings, $R = 3000$ achieves the highest Open OA among the five candidate values on both tasks. An excessively small R provides insufficient coverage of unknown samples, limiting subsequent unknown sample mining. Conversely, an excessively large R increases the likelihood of including known samples in the preliminary unknown candidate pool, reducing the reliability of subsequent supervision. A moderate R therefore provides a suitable balance between unknown sample coverage and candidate pool purity. These results support the use of $R = 3000$ as a common fixed setting across both tasks without openness-specific adjustment.

Using the above parameter settings, we further examine the performance of different OSDA baselines and our CF-CAN as the number of unknown categories increases from 1 to 5. Results are presented in Fig. 8. Most competing methods exhibit evident performance degradation or large fluctuations, particularly on open-set metrics. Among recent OSDA HSIC methods, WGDT and JDMUS show relatively competitive Open OA when the number of unknown categories is set to 4. However, their performance exhibits greater fluctuation across varying numbers of unknown categories. On the contrary, CF-CAN consistently achieves the best overall performance across all six evaluation metrics. Even when the number of unknown categories increases to 5, CF-CAN still maintains the highest Open OA at roughly 66.26%. These observations indicate that CF-CAN degrades more gracefully under increasingly challenging open-set cross-scene conditions. Such robustness primarily stems from the collaboration among the AMFF, CADT, and SSIM components. Together, they enhance representation robustness, refine classifier-level boundary calibration, and enable more reliable mining of target-domain information. As a result, CF-CAN not only remains stable with respect to key hyperparameters but also exhibits strong robustness as the number of unknown categories grows.

TABLE VII
COMPUTATIONAL COST OF DIFFERENT HSIC METHODS ON THE HOUSTON13–18 TASK

Method	D3Net [28]	FDGNet [29]	SoDa² [26]	CF-CAN (ours)
Params (M)	0.57	1.88	3.45	**0.43**
FLOPs (M)	33.83	**8.68**	202.50	25.97
Inference Time (s)	11.49	10.69	**4.04**	29.24
Open OA	70.25±1.52	67.97±2.21	54.93±0.08	**85.89±1.81**
Open AA	55.33±3.29	64.35±2.85	57.52±0.10	**80.95±1.68**
Open Kappa	44.33±0.04	48.00±0.03	36.81±0.08	**75.90±0.03**

F. Model Complexity Analysis

To assess the computational efficiency of CF-CAN, we compare it with D3Net, FDGNet, and the recently proposed SoDa² on the Houston13–18 task. The comparison considers the number of model parameters, FLOPs, inference time, and open-set classification performance. As shown in Table VII, CF-CAN contains only 0.43M parameters, the fewest among the compared methods. Its computational cost is 25.97M FLOPs, lower than those of D3Net and SoDa² but higher than that of FDGNet. Despite its compact model size, CF-CAN achieves the highest Open OA, Open AA, and Open Kappa. In

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particular, its Open OA reaches 85.89%, outperforming D3Net, FDGNet, and SoDa^2 by 15.64, 17.92, and 30.96 percentage points, respectively. CF-CAN requires a longer inference time of 29.24 s than the compared methods. This increased latency is associated with the additional operations required for dual-branch feature extraction and bidirectional feature interaction within AMFF. Nevertheless, CF-CAN delivers substantial improvements in open-set classification performance while maintaining the smallest model size and moderate computational complexity. Overall, CF-CAN provides a favorable tradeoff between open-set performance and model complexity, albeit at the cost of increased inference latency.

V. Conclusion

This article presents a unified end-to-end CF-CAN framework for open-set cross-scene HSIC. To simultaneously address the challenges of cross-scene distribution shift and the reliable discrimination between known and unknown categories, CF-CAN incorporates the AMFF module, the CADT strategy, and the SSIM module. Working in concert, these components enhance representation robustness, refine classifier-level boundary calibration, and promote more effective exploitation of target-domain information. Extensive experiments on three benchmarks consistently demonstrate that CF-CAN outperforms state-of-the-art HSIC methods. These results underscore its effectiveness and potential for practical open-set cross-scene HSIC scenarios.

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